Mixed benchmarks from earlier units Student code: _____ Calculator: yes

1. What is the value of the expression?

$$-1\frac{4}{5} \div \left(-\frac{4}{5}\right)$$

A. $\frac{4}{9}$

B. $2\frac{1}{4}$

C. $\frac{9}{20}$

D. $1\frac{11}{25}$

2. A photo is reduced by a scale factor of $\frac{1}{5}$. A side that measures 40 inches in the original is scaled down.

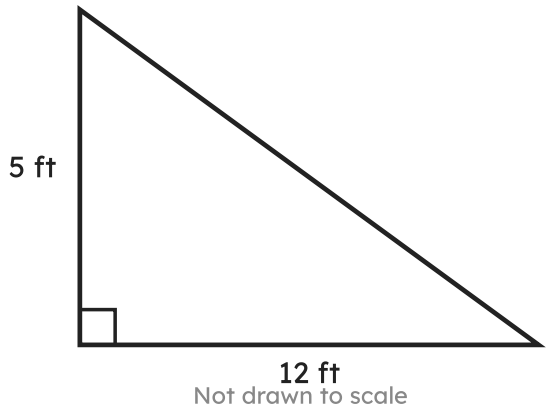
What is the length of the corresponding side in the reduction, in inches?

Turn over →

Mixed benchmarks from earlier units Student code: _____

3. A 5-foot person casts a 12-foot shadow. At the same time a tree casts a 39-foot shadow.

How tall is the tree, in feet?



Not drawn to scale

A. 7.80 ft

B. 16.25 ft

C. 93.60 ft

D. 32.00 ft

4. A student estimated the number of seats in an auditorium to be 774. The actual number was 750 seats.

What is the percent error of the estimate?

5. Write the expression using the fewest possible terms.

$$\left(\left(\frac{8}{3}\right)x - 4\right) - \left(\left(\frac{7}{3}\right)x - 5\right)$$

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — BACKSTOP-Q3

Given when no new check is ready by Thursday. Logged as a real check (Backstop = y in CHECKS).

1. MA.7.NSO.2.2 Regular **B.** $2\frac{1}{4} = \frac{9}{4}$

Dividing by a fraction means multiplying by its reciprocal.

Write the mixed number as a fraction: $-1\frac{4}{5} = -\frac{9}{5}$.

$-\frac{9}{5} \times \left(-\frac{5}{4}\right) = \frac{9}{4} = 2\frac{1}{4}$.

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2. MA.7.GR.1.5 Regular **8 inches**

Multiply by the scale factor: $40 \times \frac{1}{5} = 8$ inches.

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3. MA.7.AR.3.2 Regular **B. 16.25 ft**

The tree's shadow is $39 \div 12 = 3.25$ times as long as the person's.

The tree is 3.25 times as tall: $5 \times 3.25 = 16.25\,\text{ft}$.

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4. MA.7.AR.3.1 Regular **3.2%**

Percent error compares the size of the error with the ACTUAL amount.

Error: $|774 - 750| = 24$.

$24 \div 750 = 0.032$.

$0.032 \times 100 = 3.2\%$.

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5. MA.7.AR.1.1 Regular $\left(\frac{1}{3}\right)x + 1$

Subtracting the whole expression changes the sign of each of its terms: $\left(\frac{8}{3}\right)x - 4 - \left(\frac{7}{3}\right)x + 5$.

x-terms: $\left(\frac{8}{3}\right)x - \left(\frac{7}{3}\right)x = \left(\frac{1}{3}\right)x$. Constants: $-4 - (-5) = 1$.

$= \left(\frac{1}{3}\right)x + 1$
